Practice workbook · Improve reliability with repeats, means and larger samples

Investigations done once or on few individuals; asks how to make results more reliable, why many people are used, or how to treat an anomaly.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

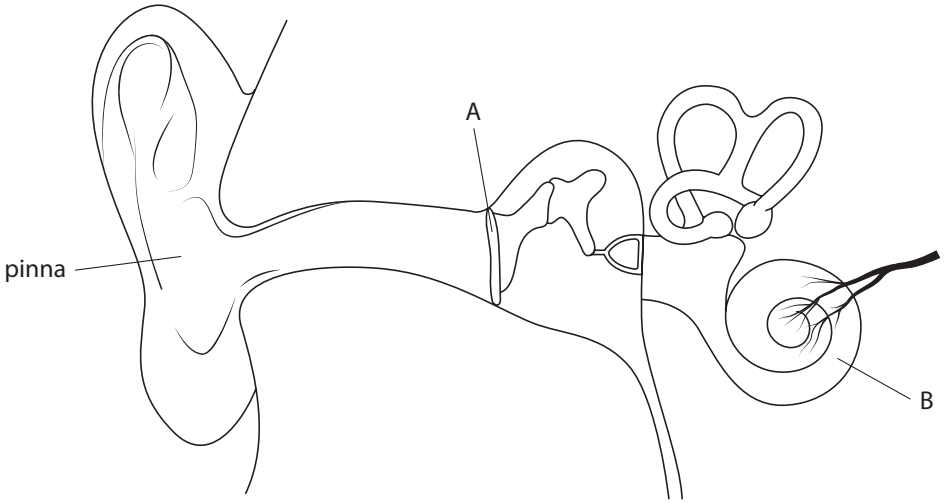
Type ID: HBI-PRAC-IMPROVE-RELIABILITY-REPEATS-SAMPLE

1. Question 5 · part c.ii · 2 marks

June 2022 Paper 2

Worked-example retry · Attempt part c.ii only.

5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 5 · part c.ii · 2 marks

June 2022 Paper 2

Worked-example retry · Attempt part c.ii only.

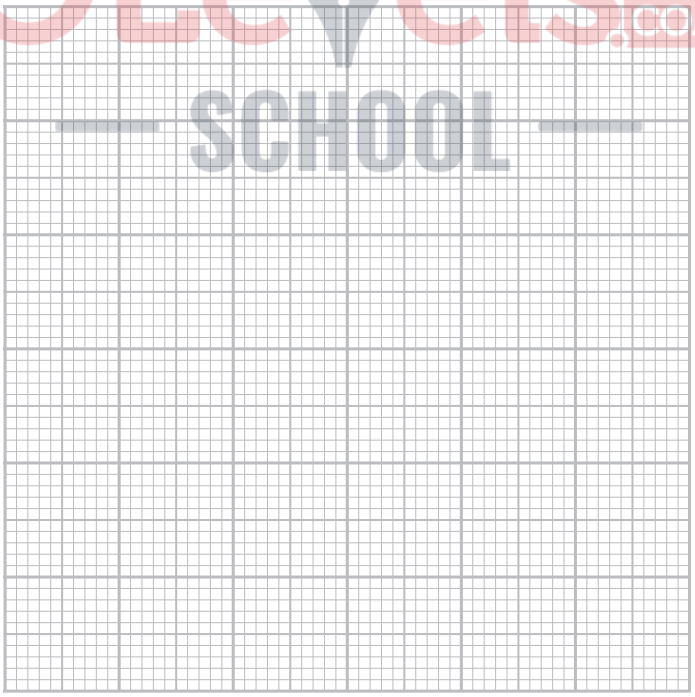
(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 0 9 7 4 A 0 1 0 2 0

1. Question 5 · part c.ii · 2 marks

June 2022 Paper 2

Worked-example retry · Attempt part c.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) Calculate the percentage change in the maximum frequency detected by people aged 20 and people aged 50.

(3)

percentage change = ... %

(c) A student suggests this method that could be used to produce the data shown in the table.

- select a person of each age to be tested
- use a signal generator to produce sounds at different frequencies
- record the maximum frequency detected by each person

(i) Give two reasons why this is unlikely to produce valid results.

(2)

1 .....

.....

2 .....

.....

(ii) Describe how the reliability of the results could be improved.

(2)

.....

.....

.....

.....

(Total for Question 5 = 15 marks)

1. Question 5 · part c.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 5 · part a.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.iii only.

5 A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

Age in years	Height of student in cm				Mean height of students in cm
	Male		Female		
12	146.2	148.9	142.8	144.2	145.5
14	165.7	166.4	161.9	164.0	164.5
16	175.9	178.5	166.3	167.6	
18	180.9	181.3	171.2	174.3	176.9

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

(2)

OLevels.com SCHOOL

mean height = ... cm

(ii) State two conclusions that the teacher could make from the data in the table.

(2)

1 .....

.....

2 .....

.....

(iii) Give two ways the teacher could improve his investigation.

(2)

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2. Question 5 · part a.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.iii only.

(b) The table shows data for two different students.

Student	Height in m	Mass in kg	BMI
X	1.69	62	21.7
Y	1.46	71	

(i) BMI is calculated using the equation

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

Calculate the BMI of student Y.
[mass measured in kg, height measured in m]

(2)

OLevels.com SCHOOL

BMI of student Y = ...

(ii) BMI can be used to assess whether a person has a healthy mass for their height.

BMI	Classification
below 18.5	underweight
18.5 – 24.9	normal weight
25 – 29.9	overweight
30 – 40	obese
above 40	morbidly obese

Using information from the table, explain how the BMI of student Y could affect her health.

(2)

.....

.....

.....

.....

(Total for Question 5 = 10 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 5 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

atropine injected →

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 2 0 5 4 A 0 1 2 2 4

3. Question 5 · part a.ii · 2 marks

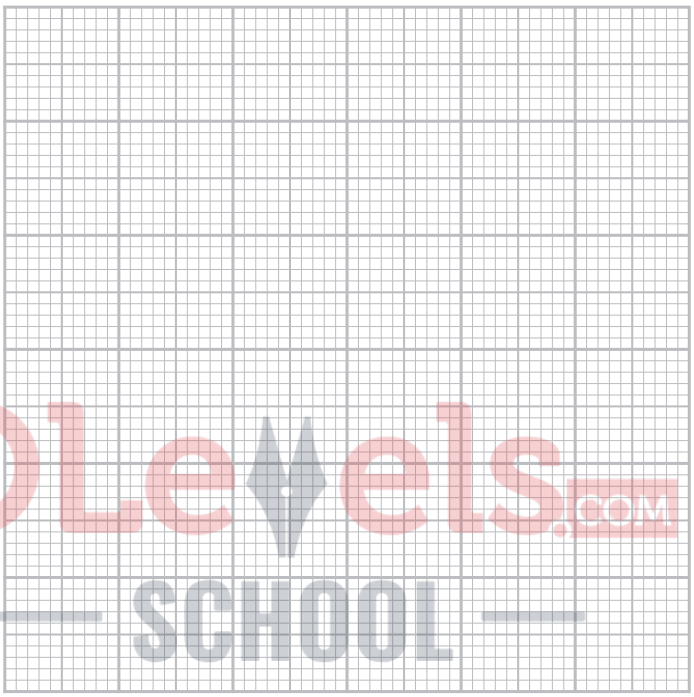
November 2020 Paper 2R

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

.....

.....

.....

.....

(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

3. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 7 · part b.iii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

OLevels.com SCHOOL

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

4. Question 7 · part b.iii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

Number of bubbles of CO_2	Time taken for chymosin to turn milk protein solid, in seconds				
	Test 1	Test 2	Test 3	Test 4	Mean
100	253	257	250	260	255
150	238	232	241	229	
200	216	214	219	211	215
250	208	202	212	198	205
300	210	200	199	311	203

(i) State three factors that the student should control.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

(2)

mean time taken = ...

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

20

P 6 2 0 5 4 A 0 2 0 2 4

4. Question 7 · part b.iii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(iii) In test 4 there is an anomalous result.

State how the student deals with this result.

(1)

.....

.....

(iv) State why recording the number of bubbles may produce inaccurate results.

(1)

.....

.....

(Total for Question 7 = 14 marks)

OLevels.com SCHOOL

Working and answer

5. Question 1 · part a.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.ii only.

1 The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 cm^3 of water, and mix into a smooth paste
- put 10 cm^3 of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

room temperature = 5 heated to 70 °C = 0

kept at 3 °C = 1 acid added = 2 alkali added = 3

no hydrogen peroxide added = 0

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 1 · part a.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) (i) Give the student's results in the form of a suitable table.

(3)

(ii) State why this investigation is not reliable.

(1)

(iii) Give a reason why the results of this investigation are not accurate.

(1)

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

(3)

OLevels.COM — SCHOOL —

5. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 2 · part b.iii · 2 marks

June 2023 Paper 2R

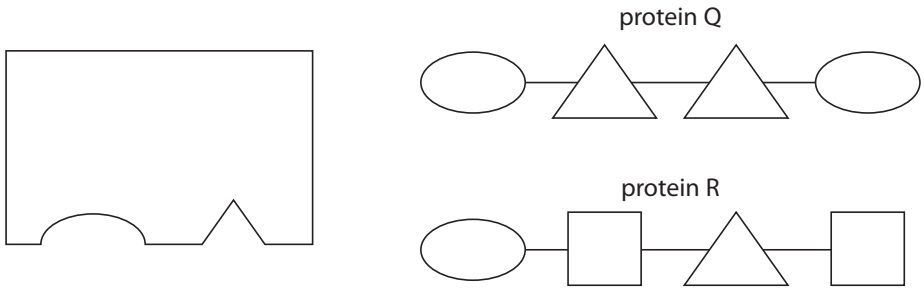
Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

2 (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.



Enzyme P **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

(3)

amino acids	amylases	fatty acids	proteases	protein Q	protein R	starch	sugar

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

(1)

.....

(iii) Name the molecule that codes for the production of enzyme P.

(1)

.....

OLevels.COM SCHOOL

6. Question 2 · part b.iii · 2 marks

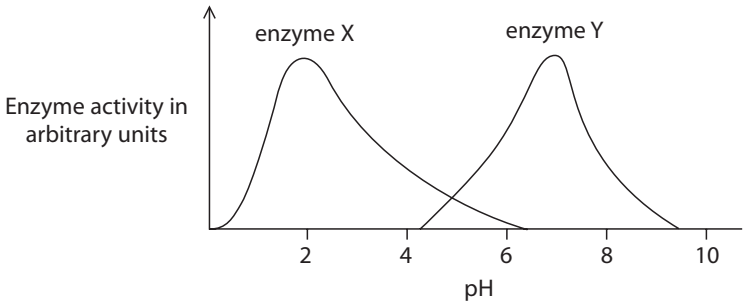
June 2023 Paper 2R

Further practice · Attempt part b.iii only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.



(i) Describe how the activity of enzyme X changes with pH.

(3)

(ii) Explain why no activity is shown for enzyme Y at pH2.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 2 5 8 8 A 0 6 2 4

6. Question 2 · part b.iii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(iii) Describe how the student could make the results of this investigation more reliable.

(2)

.....

.....

.....

.....

(iv) Name one control variable in the student's investigation.

(1)

.....

(Total for Question 2 = 13 marks)

OLevels.com SCHOOL

Working and answer

7. Question 4 · part b.i · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4 (a) (i) Complete the word equation for one type of respiration.

(2)

oxygen + ... → ..... + ...

(ii) State the type of respiration shown by the equation.

(1)

.....

(iii) In which part of the cell does this type of respiration occur?

(1)

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

(1)

.....

OLevels.com SCHOOL

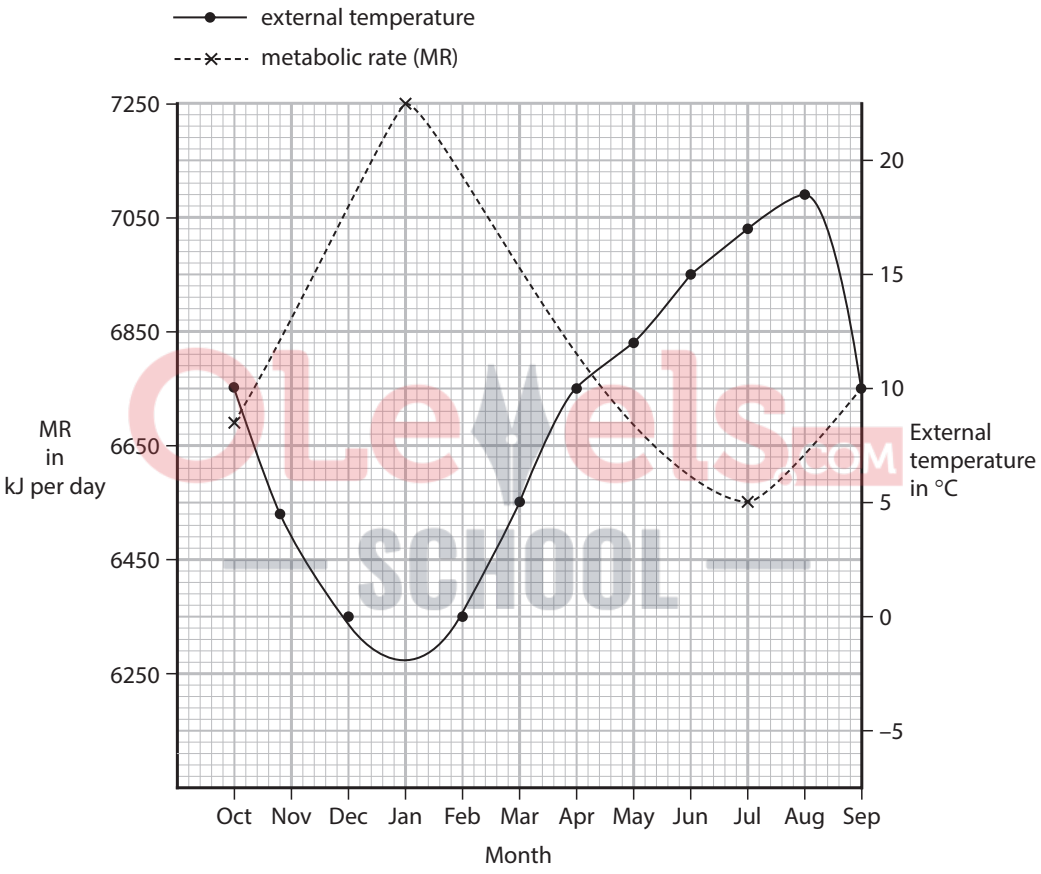
7. Question 4 · part b.i · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.i only.

(b) An investigation was carried out to see how a person's metabolic rate (MR) and the external temperature change at different times in one year.

The graph shows the results of the investigation for one person.



(i) Describe how the investigation could be made more reliable.

(2)

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

10

P 7 2 5 8 7 A 0 1 0 2 4

7. Question 4 · part b.i · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.i only.

(ii) Determine the difference between the maximum metabolic rate and minimum metabolic rate.

(2)

difference = ... kJ per day

(iii) Explain why the metabolic rate changes throughout the year.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 2 · part e · 1 mark

November 2024 Paper 1

Further practice · Attempt part e only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

OLevels.COM SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8. Question 2 · part e · 1 mark

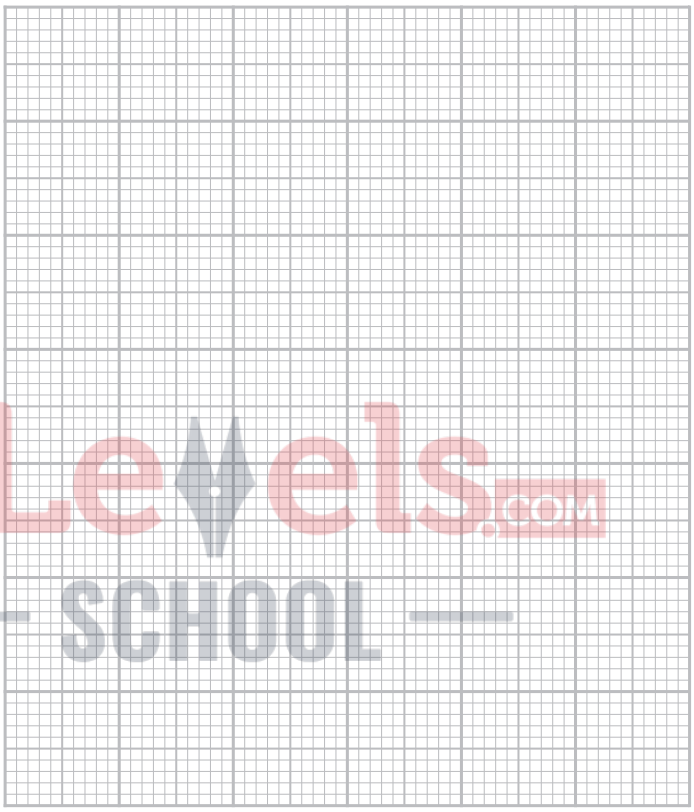
November 2024 Paper 1

Further practice · Attempt part e only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(a) (i) Plot a graph of the student's results.

(3)



olevels.com SCHOOL

(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

.....

.....

8. Question 2 · part e · 1 mark

November 2024 Paper 1

Further practice · Attempt part e only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

(3)

(c) Name the independent variable in this investigation.

(1)

(d) Give two variables that the student should control in this investigation.

(2)

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

(1)

(Total for Question 2 = 12 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 2 · part e · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 7 · part d · 2 marks

June 2021 Paper 2

Further practice · Attempt part d only.

7 Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 cm^3 of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 cm^3 of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

Petri dish	Diameter of bacterial growth in mm				
	day 1	day 2	day 3	day 4	day 5
A	0	1.7	3.0	4.6	7.1
B	0	4.2	8.4	15.1	36.5
C	0	0	0	0	0

(a) (i) State three control variables in the student's investigation.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

9. Question 7 · part d · 2 marks

June 2021 Paper 2

Further practice · Attempt part d only.

(ii) State the dependent variable in the student's investigation.

(1)

.....

(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

(2)

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

(2)

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

O Levels .COM — SCHOOL —

9. Question 7 · part d · 2 marks

June 2021 Paper 2

Further practice · Attempt part d only.

(d) Suggest what further tests the student should do to show that garlic can be used as an effective general antibacterial agent.

(2)

DO NOT WRITE IN THIS AREA

(Total for Question 7 = 14 marks)

OLevels.com SCHOOL

Working and answer